

SANCHIT GOYAL

BACKEND & DISTRIBUTED SYSTEMS ENGINEER | TECH LEAD

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PROFESSIONAL SUMMARY

Tech Lead with 10+ years in backend engineering, currently at PayPay — Japan's largest mobile payment service, 75M registered users. Owns the multi-tenant communications platform behind four PayPay group financial entities, delivering 35M notifications/day at 20,000/second peak on 99.99% uptime, and sets technical standards and design review across 7–8 engineers. Owned a 25 TB, ~100-billion-row TiDB-to-Aurora migration that cut \$1M in annualized cost, and took an LLM customer-support agent from PoC to 100% rollout. Deep in Java, Kotlin, Python, Spring Boot, Kafka, AWS, and distributed systems for money movement.

CORE SKILLS

Languages & Frameworks: Java, Kotlin, Python, Spring Boot

Data & Infrastructure: Kafka, Aurora, MySQL, TiDB, Cassandra, Redis, AWS, Kubernetes, Docker, Terraform, Prometheus, Grafana

Architecture: Distributed Systems, Microservices, Event-Driven Architecture, API Design, Concurrency, Multi-Tenant Platforms, System Design, Live Data Migration, Observability, High Availability

Domain: Payments, Digital Wallets, Ledger & Reconciliation, Idempotency, Regulated Financial Systems

AI & ML: LLM Applications, RAG, AI Agent Design, Agent Memory

PROFESSIONAL EXPERIENCE

Tech Lead | PayPay India

Gurgaon, India | Apr 2024 – Present

PayPay — Japan's largest mobile payment service; 75M registered users, Nasdaq-listed

- Owned a multi-tenant communications platform for PayPay, PayPay Card, PayPay Bank, and PayPay Securities, sustaining 35M notifications/day at 99.99% uptime; created sister-company onboarding guidelines and isolated tenants to prevent noisy-neighbor impact.
- Scaled push, email, SMS, in-app, LINE, WhatsApp, and merchant-webhook channels 5×, from 4,000 to 20,000 notifications/second, using Kafka/queues, worker autoscaling, batching, rate limiting, and tenant quotas/isolation — capacity sized for Japan's restricted promotional sending window, which compresses the day's campaign volume into a few hours, with same-day campaigns due before noon.
- Built end-to-end observability on Prometheus and Grafana that connects notifications to campaigns or transactions, enabling CTR/CVR measurement for campaigns and SLA monitoring for transactional sends, delivered within 5 seconds of payment. Diagnosed and shipped a fix for a Java 21 virtual-thread issue within one week.
- Authored platform design documents and set the technical standards four sister companies onboard against; reviewed and approved designs for 7–8 engineers.
- Launched PayPay Points (Limited-Time) to production, owning point-system implementation and reconciliation for expiry-aware, restricted-service points. Solved complex refund/reconciliation flows, supported 350 TPS peak point grants, and received a Rockstar Award.
- Owned a 25 TB, ~100-billion-row TiDB-to-Aurora migration from PoC through production release, partnering with two DBAs — 75M wallets, ~225M accounts, and four ever-growing transaction and idempotency tables. Dual-write replication kept Aurora current for a 30-minute cutover window; live partition rotation and archival contain growth against Aurora's 64 TB per-table ceiling. Cut \$1M in annualized cost.
- Designed the RAG knowledge pipeline and agent-memory system for an LLM customer-support agent; took it from PoC through pilot to 100% rollout, with P90 time-to-first-token of 5 seconds and cost below ¥5 (~US\$0.03) per conversation. Implemented public-knowledge grounding and human escalation for high-risk financial issues; technically guided 3–4 engineers on the agent workstream.

Senior Backend Engineer / PayPay India

Gurgaon, India / Oct 2023 – Mar 2024

- Contributed to migrating PayPay Point accounts from a complex wallet domain into a dedicated point system. Under an architect-defined HLD, a seven-engineer team owned LLD, cross-team coordination, implementation, and phased rollout.
- Delivered data/API migration, reconciliation, testing, and production-readiness work; completed a 100% phased rollout with no data-consistency incidents. Raised supported peak throughput from 750 to 1,000 TPS and cut P99 payment latency from 150ms to 100ms, with end-to-end payment P99 down from 1.2s to 0.9s.

Senior Lead Engineer / Guardian India Operations

Gurgaon, India / Aug 2021 – Sep 2023

Guardian Life — US Fortune 250 insurance & healthcare group

- Led and mentored 5+ engineers modernizing legacy healthcare platforms into microservices, including phased service decoupling and migration to a new claims platform.

Lead Engineer / Guardian India Operations

Gurgaon, India / Jan 2020 – Jul 2021

- Designed cloud-to-legacy integration architecture and delivered an Angular self-service configuration application on AWS Lambda, API Gateway, and DynamoDB, saving \$12K per client annually; integrated Drools business rules to reduce client unloading time by 50%.

Software Engineer / Orange Business Services

Gurgaon, India / Aug 2019 – Dec 2019

Global network & digital services arm of Orange S.A. (France)

- Built four reusable components and scalable application features, reducing estimated development effort by 20% across multiple projects.

Business Systems Analyst / Optum Global Solutions

Gurgaon, India / Aug 2018 – Jul 2019

UnitedHealth Group — the world's largest healthcare company

- Designed architecture for a microservices migration and developed a cloud-based automation proof of concept; automated manual work to eliminate approximately 20 hours per week and improve core processes 3×.

Software Engineer / Optum Global Solutions

Gurgaon, India / Jan 2018 – Jul 2018

- Modernized a framework to reduce processing time by 3×.

Associate Software Engineer / Optum Global Solutions

Gurgaon, India / Jul 2016 – Dec 2017

- Delivered features and production fixes in Java, JavaScript, and jQuery.

EDUCATION

B.Tech, Electronics & Communications Engineering | Greater Noida Institute of Technology (Dr. A.P.J. Abdul Kalam Technical University), India | 2016